



Shandong Penglai Pingdingshan Wind Farm Project



Document Prepared By Juno Carbon Investment & Environment Technology (Beijing) Co., Ltd.

Project Title	Shandong Penglai Pingdingshan Wind Farm Project
Version	2.1
Report ID	-
Date of Issue	11-November-2022
Project ID	#258
Monitoring Period	26-October-2011 to 31-December-2016
Prepared By	Juno Carbon Investment & Environment Technology (Beijing) Co., Ltd.
Contact	Ding Xue +86 186 0095 3503 http://www.junocapital.com.cn/ dingxue@junocapital.cn Address:1701, Building 2, Yard A 2, Jiangtai Road, Chaoyang District, Beijing

CONTENTS

1 PROJECT DETAILS	4
1.1 Summary Description of the Implementation Status of the Project.....	4
1.2 Sectoral Scope and Project Type	5
1.3 Project Proponent	5
1.4 Other Entities Involved in the Project.....	5
1.5 Project Start Date	5
1.6 Project Crediting Period	5
1.7 Project Location	6
1.8 Title and Reference of Methodology	7
1.9 Participation under other GHG Programs	8
1.10 Other Forms of Credit	8
1.11 Sustainable Development Contributions	9
2 SAFEGUARDS	11
2.1 No Net Harm	11
2.2 Local Stakeholder Consultation	12
2.3 AFOLU-Specific Safeguards	14
3 IMPLEMENTATION STATUS	14
3.1 Implementation Status of the Project Activity	14
3.2 Deviations	15
3.3 Grouped Projects	16
4 DATA AND PARAMETERS	16
4.1 Data and Parameters Available at Validation	16
4.2 Data and Parameters Monitored	16
4.3 Monitoring Plan	23

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	28
5.1 Baseline Emissions	28
5.2 Project Emissions	28
5.3 Leakage	28
5.4 Net GHG Emission Reductions and Removals	29
APPENDIX 1: < MONTHLY AND YEARLY ELECTRICITY SUPPLY >	31
APPENDIX 2: < EVIDENCE OF ACHIEVED SD CONTRIBUTIONS >	44

1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Shandong Penglai Pingdingshan Wind Farm Project (hereafter refers to “the project”) developed by China Resources Wind Power (Yantai) Co., Ltd. is a grid-connected renewable energy Project in Beigou Town, Penglai City, Shandong Province, P.R. China.

The Project involves the installation of 15 sets wind turbines with unit capacity of 2MW, and 10 sets wind turbines with unit capacity of 1.8MW, which totals up an installation capacity of 48MW, with a predicted power supplied to the grid of 101,904 MWh per year, and the annual full-load operation time amount to 2,123h.

The purpose of the Project is to utilize a wind power facility to generate zero greenhouse gas (GHG) emissions electricity for the North China Power Grid (NCPG) thereby displacing electricity that is relatively carbon intensive. The project is therefore expected to reduce emissions of GHG by an estimated 107,488 tCO₂e per year during the first crediting period (20/12/2008-19/12/2018) by displacing electricity from the Grid

The scenario existing prior to the start of the implementation of the project is: The same electricity output by the project activity would have otherwise been generated by the operation of NCPG connected power plants and by the addition of new generation sources. That is the same as the baseline scenario.

The Project started to construct on 25/04/2008. It started commissioning on 20/12/2008, and was put into full delivery on 21/01/2009.

The project has been registered as a CDM project on 21/07/2009, and the reference number is 2397¹. CERs of 245,407 tCO₂e have been issued for the monitoring period from 21/07/2009 to 25/10/2011.

The project has been registered as a VCS project with VCS ID of 258 and VCUs of 71,328 tCO₂e have been issued for the monitoring period from 20/12/2008 to 20/07/2009. This VCS monitoring period covered the time from 26/10/2011 to 31/12/2016² and the emission reductions in this monitoring period are 631,397 tCO₂e.

¹ <https://cdm.unfccc.int/Projects/DB/RWTUV1234176728.19/view>

² CDM-MR during the 4th monitoring period from 26/10/2011 to 24/06/2012 have been submitted to EB for publication but the request for issuance has not been submitted and the verification contract between PP and VVB (BV) has been terminated, the project owner will also not apply for the issuance of CERs again. Furthermore, on 15/08/2022, the monitoring report withdrawal request form has been submitted to EB. And PP promises that the emission reduction in this monitoring period will only be issued under the VCS programs.

1.2 Sectoral Scope and Project Type

This category would fall within sectoral scope 1: energy industries (Renewable/non-renewable sources).

Project type: wind power project.

The project is not a grouped project.

1.3 Project Proponent

Organization name	China Resources Wind Power (Yantai) Co., Ltd.
Contact person	Zhao Jie
Title	Project Manager
Address	Floor 25th, Luneng International Center, No.2666, Erhuan Road on the West, Shizhong District, Jinan 100015, China.
Telephone	+86 139 5408 2518
Email	Zhaojie192@crpower.com.cn

1.4 Other Entities Involved in the Project

Organization name	Juno Carbon Investment & Environment Technology (Beijing) Co., Ltd.
Role in the Project	VCUs buyer
Contact person	Ding Xue
Title	Technical Director
Address	1701, Building 2, Yard A2, Jiangtai Road, Chaoyang District, Beijing
Telephone	+86 186 0095 3503
Email	dingxue@junocapital.cn

1.5 Project Start Date

This project starts commissioning on 20/12/2008, which means the date that started to generate GHG emissions.

1.6 Project Crediting Period

20/12/2018-19/12/2018 (renewable crediting period which covers 10 years).

Moreover, the project is also a registered CDM project (Ref. 2397) with a 7*3 renewable project crediting period starting on 21/07/2009, the VCS issuance is not eligible for beyond the end of those 21 years, and the project's expected operation lifetime is 20 years in the registered CDM-PDD. Thus, given that the VCS crediting period started on 20/12/2008, and the project has fail to renew the crediting period within two years after the end of the first project crediting period, the project proponent confirms the VCS crediting period of the project activity will end on 19/12/2018.

This monitoring period is from 26/10/2011 to 31/12/2016 within the first VCS crediting period and the monitoring period covers 1,893 days.

The Project is located in Beigou Town, Penglai City, Shandong Province, People's Republic of China. The geographical coordinates of the Project are east longitude of 120°38'48"~120°42'11" and north latitude of 37°44'09"~37°46'17". The nearest wind turbine to the substation is No.11, with east longitude of 120°42'01"and north latitude of 37°46'14".

Figure 1 and 2 below show the geographical location of the project.



Figure 1. The project on the map of P. R. China

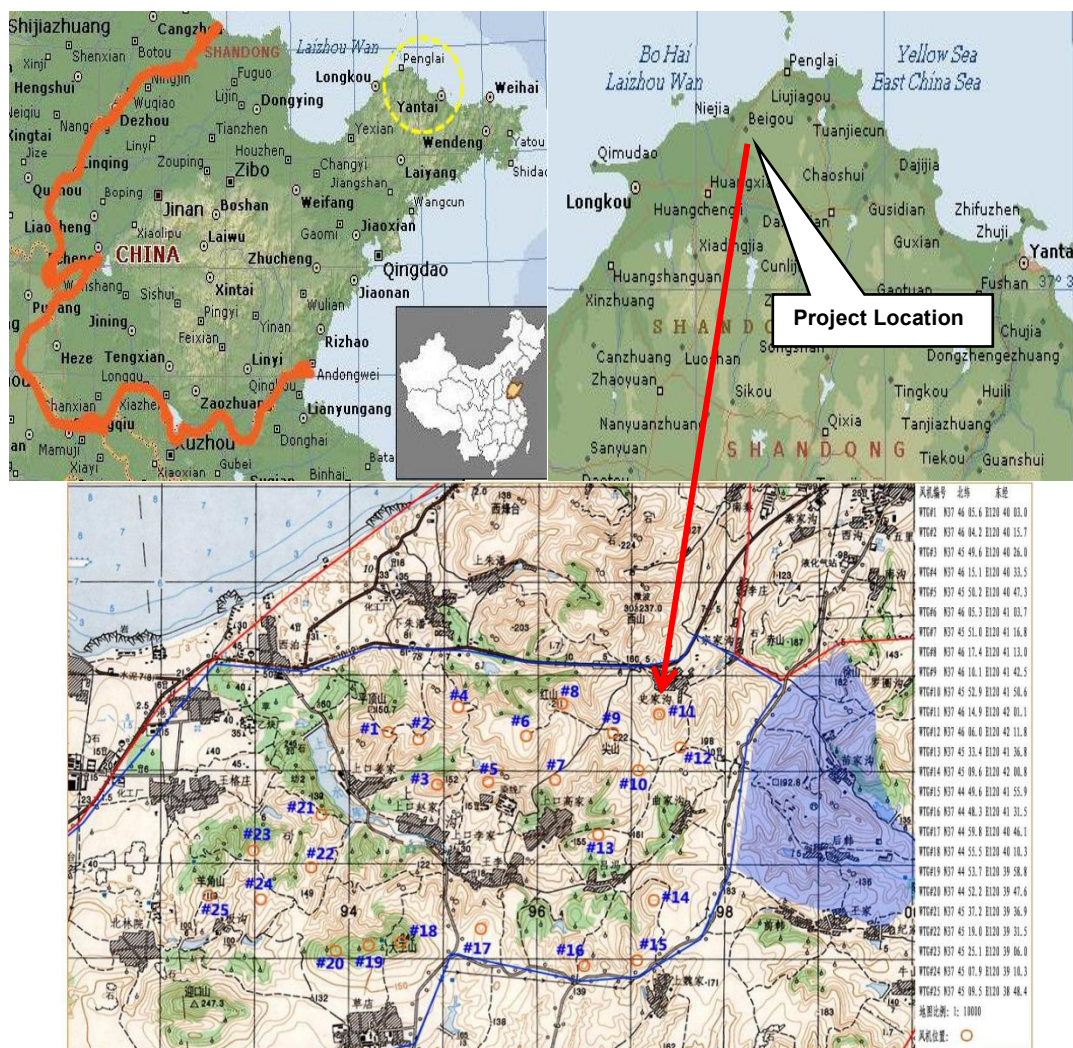


Figure 2. The project on the map of Penglai City, Shandong Province, P. R. China

1.8 Title and Reference of Methodology

The approved methodology applied in the project is ACM0002 (version 07 dated 30/11/2007) – “Consolidated baseline and monitoring methodology for grid-connected electricity generation from renewable sources”.

The approved methodology “Tool for the Demonstration and Assessment of Additionality, (version 05.2 dated 26/08/2008)” and “Tool to calculate the emission factor for an electricity system, (version 01.1 dated 29/07/2008)” are applied in the project.

For more information regarding the methodologies please refer to:

<https://cdm.unfccc.int/methodologies/index.html>

1.9 Participation under other GHG Programs

The project has been registered as a Clean Development Mechanism (CDM) project in UNFCCC on 21/07/2009 (UNFCCC Ref. 2397), with the first 7 years crediting period started from 21/07/2009. CERs of 39,987 tCO₂e have been issued for the 1st monitoring period from 21/07/2009 to 24/11/2009. CERs of 134,695 tCO₂e have been issued for the 2nd monitoring period from 25/11/2009 to 24/02/2011. CERs of 70,725 tCO₂e have been issued for the 3rd monitoring period from 25/02/2011 to 25/10/2011. The project has been registered as a VCS project with Ref. VCS 258 on 20/12/2008 and VCUs of 71,328 tCO₂e have been issued for the monitoring period from 20/12/2008 to 20/07/2009.

All credits from 26/10/2011 to 31/12/2016 will be claimed under VCS program as VCUs for the project to avoid double counting.

The issuance status is shown as below:

Monitoring period		Status	CERs/VCUs (tCO ₂ e)
1 st Monitoring period under VCS	20/12/2008~20/07/2009	Issued	71,328
1 st Monitoring period under CDM	21/07/2009~24/11/2009	Issued	39,987
2 nd Monitoring period under CDM	25/11/2009~24/02/2011	Issued	134,695
3 rd Monitoring period under CDM	25/02/2011~25/10/2011	Issued	70,725

1.10 Other Forms of Credit

- Emission Trading Programs and Other Binding Limits

The project has been registered as a CDM project on 21/07/2009 (Ref. 2397) and CERs of 245,407 tCO₂e have been issued for the 1st-3rd monitoring period from 21/07/2009~25/10/2011.

The project proponent China Resources Wind Power (Yantai) Co., Ltd. is an enterprise for renewable energy investment, is not included in the compliance entity list by China national Emission Trading Scheme (ETS). Also, the project has not been registered as a CCER (Chinese Certified Emission Reductions) project in China, thus it is not eligible for emission reductions transaction under the China ETS. Therefore, the project is not involved in an emission trading program or any other mechanism that includes GHG allowance trading. Furthermore, as a voluntary emission reduction project, the GHG emission reductions and removals generated by the project will not be enforced to be used for compliance in China.

Furthermore, as declared by the project proponent the credits in this monitoring period (26/10/2011-31/12/2016) has not been counted and will not be counted under other kind of GHG programs.

- Other Forms of Environmental Credit

The project hasn't sought or received another form of environmental credits.

1.11 Sustainable Development Contributions

The project makes contribution to the sustainable development as follows:

The project produces clean renewable energy by the utilization of wind power to diminish CO₂ emissions, and creates direct and indirect employment opportunities during construction and operation phases.

During this monitoring period, 598,594.662 MWh of electricity from renewable sources has been exported to the power grid which contributes to SDG 7: Affordable and Clean Energy³ and China's National Plan on Implementation of the 2030 Agenda for Sustainable Development⁴ 7 through increasing the share of clean energy consumption.

During the construction, operation, and maintenance of this project, the project, directly and indirectly, generates more job opportunities, which helps improve local employment and reduce local poverty. Thus, the project achieved SDG 8: Decent Work and Economic Growth⁵ and China's National Plan on Implementation of the 2030 Agenda for Sustainable Development 8: Increase labor force participation rate through implementation of the classification policy. Vigorously enforce the Law on Promotion of Employment.

And the project has achieved a GHG emission reduction of 631,397 tCO₂e during this monitoring period. Thus, the project achieved SDG 13 Climate Action⁶ and China's National Plan on Implementation of the 2030 Agenda for Sustainable Development 13: Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas.

These contribute to achieve China's stated sustainable development priorities, included strengthening resistance capacity to climate risks, increasing employment rate, increasing the share of clean energy consumption and reducing emissions of GHG. Refer to table 1 for detail. The evidences of the project's SD contributions during the first VCS monitoring period (20/12/2008-20/07/2009) and this monitoring period (26/10/2011-31/12/2016) have been provided in the Appendix 2 of this report.

³ <https://sdgs.un.org/goals/goal7>

⁴ <http://www.gov.cn/xinwen/2016-10/13/5118514/files/44cb945589874551a85d49841b568f18.pdf>

⁵ <https://sdgs.un.org/goals/goal8>

⁶ <https://sdgs.un.org/goals/goal13>

Table 1: Sustainable Development Contributions⁷

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7.2.1 Renewable energy share in the total final energy	Implemented activities to increase	The project has utilized wind energy for electricity generation, the electricity supplied from the project is 666,216.962 MWh.	The project has utilized wind energy for electricity generation, the electricity supplied from the project is 666,216.962 MWh which contribute the SDGs 7.
2)	8.5	Number of jobs created	Implemented activities to increase	24 people including 5 females were employed by the project owner for the operation, monitoring and equipment maintenance of this project, which helps increase the employment rate.	The project activity had provided temporary working opportunities during the construction period. From the operation start date, 24 people were employed including 5 females during the first VCS monitoring period and this monitoring period which contribute the SDGs 8.
3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	By utilizing wind for power generation and replaced electricity from thermal power generation projects, the project has supplied clean energy to NCPG of 666,216.962 MWh and has avoided 702,725 tCO ₂ e release into the atmosphere.	The project has prevented the release of 702,725 tCO ₂ e into the atmosphere during the first VCS monitoring period and this monitoring period which contribute the SDGs13.

⁷ Current project contributions and contributions over project lifetime in this table including both the contribution during the first monitoring period (20/12/2008-20/07/2009) and this monitoring period (26/10/2011-31/12/2016) under VCS.

2 SAFEGUARDS

2.1 No Net Harm

The project uses wind to generate electricity. It is environmentally friendly and there is no harm to the environment. The Environmental Impact Assessment (EIA) for the project was completed in 09/08/2006 by Yantai Institute of Environmental Protection Science, and was approved by Shandong Provincial Environment Protection Bureau on 10/09/2007. A summary of the EIA is presented below.

Noise

Noise is one of the main adverse effects of wind power generation on the environment. It comes from the stable and continuous low-frequency noise generated by blade cutting airflow and the noise generated by internal mechanical operation. In this project, the low-noise fan is selected, and the vibration absorption system is installed at the connection of the fan equipment. The blades are made of sound absorbing materials, which significantly reduces the noise compared with the old-style fan.

Ambient air

After the project is put into operation, it has no adverse effect on the ambient air, and belongs to the "green industry" encouraged by the state. Electric heating equipment is used for winter heating, and liquefied gas or electric furnace equipment is used for the life and diet of wind farm workers, which has little impact on regional ambient air quality.

Sewage

The sewage generated by the project mainly comes from the daily life of the personnel on duty, and the output is small. After being treated by sewage treatment facilities, it is used to water trees, which has little impact on the environment.

Solid waste

In addition to the regular maintenance of the fan generated a small amount of waste oil and domestic garbage, no other solid waste generated. During the operation period of the project, waste oil and packaging materials are collected uniformly, which will not have adverse impact on the environment.

Light pollution and electromagnetic radiation

Because the minimum distance between the fan and the residential area is 300 meters, the terrain of the proposed project area is flat, the height difference between the village and

the fan is not large, and the fan blades are treated with matte light to reduce light pollution and not cause light pollution to the village.

There are no residential areas and wireless communication equipment within 300m around the wind farm in this period, so it can be considered that the wind farm has little interference to local residents, radio and television.

Ecological impact

The construction process of the project destroyed a certain amount of surface vegetation, the project through the fan foundation around the road and both sides of the afforestation, to compensate and restore the ecological environment. In conclusion, the construction work of this project will not have an impact on the local ecological environment.

With mitigation controls planned as part of the Project construction and EIA process, and the contribution made by the project to sustainable development for the local and national area, i.e. providing job opportunities to the local community, the Project is expected to have an overall positive impact on the local and global environments. Mitigation measures will ensure that there are no significant residual impacts associated with the Project.

2.2 Local Stakeholder Consultation

LSC prior to the project implementation

On 17 February 2008, the project owner carried out a survey of the local residents around the project location. The consultation invited local stakeholders to submit comments on the Project activity by filling in a questionnaire sent out by the Project Developer. The staff introduced the background of the proposed project and then sent out 50 copies of questionnaire in a random way, 50 pieces of reply were received. Among the interviewees, 31 of them are farmers, 10 are workers, 8 are governmental officials and 1 student.

An invitation notice for stakeholder comments was later issued by the project developer on March 1st, 2008 and 19 representatives of local stakeholders, including governmental officials of local County and local residents, etc. attended the meeting on March 11 to discuss the questionnaires collected and further introduce the project. No negative opinion on construction of the project is heard and environmental considerations expressed by stakeholders are discussed on the meeting.

The questions regarding the proposed project were mainly as follows:

- Is the current living and/or working environment quiet?
- Do you currently experience electromagnetic interference when watching TV at home?
- Are there any negative impacts of the proposed project on the everyday life of local residents?

- Is the proposed project going to help improve the living and/or working environment?
- Which is the environmental topic that concerns you the most during the construction and operation of the proposed project?
- Do you support the proposed project?

The summary of questionnaire survey is listed as the following:

- 50 (100%) of them consider their current living and/or working environment is quiet;
- 48 (96%) of them currently do not experience electromagnetic interference when watching TV at home and 2 (4%) of them is unsure;
- 50 (100%) of them think there will not be any negative impacts on their everyday life;
- 49 (98%) of them think the proposed project will help improve their living and/or working environment, while 1 (2%) of them is unsure;
- Regarding the construction and operation of the proposed project, 13 (26%) of them are most concerned with electromagnetic interference, 33 (66%) of them are most concerned with the noise level, and 4 (8%) of them are most concerned with wastewater from the project;
- 50 (100%) of them support the implementation of the proposed project.

The summary of Stakeholder meeting:

- The stakeholder meeting discussed the questionnaires received, from which it shows the local community possesses basically positive comments on the effects of the proposed project. They believe that there will be improvement for local economic development. At the same time, they also express consideration about potential impact on environment. As to noise, the distance between a local plant to one of the original wind turbine locations is very near, considering the noise interruption in the future, the turbine has been moved to other place, thus to mitigate the noise.

LSC during the operation period

To keep on-going communications with local stakeholders, the project owner has public its phonenummer of the central control room at the power plant gate and also oral noticed to local people and village committees. Anyone who have comments on the project could phone the project owner directly. Besides, the project owner also put a grievance book in the office of wind plant and the villages can leave their grievance or suggestions on the book anytime.

Meanwhile, the local authority has also conducted spot checks on the implementation of the project from time to time as per the request from the local governments' regulations.

There are no negative comments or grievance received for the project. In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

2.3 AFOLU-Specific Safeguards

The project is not AFOLU project.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The Project involves the installation of 25 sets of wind turbines, of which 15 sets with a unit capacity of 2 MW capacities and 10 sets with a unit capacity 1.8, which totals up an installation capacity of 48MW.

The wind turbines are supplied by Vestas, with the model of V90-2.0MW and V90- 1.8MW. The expected full-load operation time amount to 2,123h per year and the estimated net annual power supplied to the grid is 101,904 MWh.

The V90-2.0MW and V90-1.8MW turbines are optimized for site with low turbulence and low and medium winds. The Opti Speed⁸ generators present a significant advance in wind turbine efficiency, which helps to boost annual power generation. The main technical specifications of the wind turbines are provided in the following table.

Table 2: key technical specifications of the wind turbines

Parameter		Data	
Model		V90-2.0MW	V90-1.8MW
Quantity		15	10
Height of hub		80 m	
Wind Turbine	Diameter	90 m	
	Nominal revolutions	14.9 rpm	
	Number of blades	3	
	Nominal wind speed	13 m/s	12 m/s
	Cut-in wind speed	3 m/s	3 m/s

⁸ Opti Speed allows the rotor speed to vary within a range of approximately 60 per cent in relation to normal rpm. Thus with Opti speed, the rotor speed can vary by as much as 30 per cent above and below synchronous speed.

Parameter		Data	
	Cut-off wind speed	25 m/s	25 m/s
	Power regulation	Pitch/Opti Speed	
Generator	Rated Power	2.0MW	1.8MW
	Rated voltage	690V	
	Life time	20 Years	

A 220 kV substation has been set up at the site. The wind power generated is switched through 220 kV substations at the project site, and then connected to the 220 kV Tangqiu substation, and then transmitted to the North China Power Grid finally.

The starting date of the construction was 25/04/2008. The project started commissioning on 20/12/2008 and was put into full delivery on 21/01/2009.

During this monitoring period (from 26/10/2011 to 31/12/2016), the monitoring activities were conducted strictly in accordance with the approved revised monitoring plan. The project has operated smoothly without any accidental or emergency events that might impact the accuracy and/or implementation of monitoring activities

3.2 Deviations

3.2.1 Methodology Deviations

There is no methodology description deviation in this monitoring period.

3.2.2 Project Description Deviations

In the registered VCS PD, the crediting period is described as from 20/12/2008 to 20/07/2009. A deviation is requested for the crediting period in the registered VCS PD. The project is registered under VCS Standard 2007 and completed validation before 19/03/2020. Thus, it remains eligible to apply the crediting period requirements under VCS 2007 which shall be a maximum of ten years and may be renewed at most twice, so the first renewable crediting period of the project should be updated from 20/12/2008 - 20/07/2009 to 20/12/2008-19/12/2018, and it has no impacts on the applicability, additionality or baseline of the project.

After the first CDM monitoring period (from 21/07/2009 to 24/11/2009) of the Project, it was noticed that the Project shared the same 220kV substation (Tangqiu 220kV Substation) with the other wind farm, namely Shandong Penglai Wind Farm Phase II Project (hereafter refers to Project B), which was put into operation on 25/11/2009. The main gateway meter (M1) installed at the Tangqiu 220kV Substation monitors the electricity supplied to and imported from the grid by the Project and Project B. Separate meters (M4, M5, M6, M7 and M8) have been installed at each project site to measure electricity supplied. The grid company

calculates the net electricity supplied by the Project based on the meter readings of the gateway meter (M1) and these on-site separate meters (M7, M8) of Project B.

Considering the actual monitored parameters have not been specified in the monitoring plan in the registered PDD. The monitoring plan of the Project has been revised to including actual monitoring parameters. The revised monitoring plan was approved by EB on 24/02/2011.

There is no other project description deviation in this monitoring period.

3.3 Grouped Projects

The project is not a grouped project.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ e/MWh
Description	Baseline emission factor of NCPG in the monitoring period.
Source of data	The registered PDD.
Value applied	1.0548
Justification of choice of data or description of measurement methods and procedures applied	Ex-ante determined in the registered PDD and fixed during the 1 st crediting period.
Purpose of Data	Calculation of baseline emission.
Comments	Other ex-ante parameters (i.e. FC _{i,m,y} , NCV _{i,y} , EF _{CO2,i,y} , Carbon Oxidation Factor, Installed Capacity, GEN _{import,y} , GEN _y and Efficiency level of the best technology commercially available for coal, oil and gas-fired power plant) were used to calculate the parameter EF _{grid,CM,y} during the validation stage.

4.2 Data and Parameters Monitored

Data / Parameter	EG _{export,project}
Data unit	MWh
Description	Electricity supplied by the project activity to the grid
Source of data	Project activity site with electricity meters M4, M5, and M6 installed on the collection line 1, 2, and 3.
Description of	Electricity supplied to the grid by the project activity is monitored

measurement methods and procedures to be applied	through the bi-directional electricity meters (M4, M5 and M6). The meters (M4, M5 and M6) are installed on the collection lines 1, 2 and 3 of the project activity. The accuracy of above meters is 0.2S. The records of the meters are based on continuous measurement and monthly recorded by the project owner. The cut-off time of recording of all meters is at 24:00 on the last day of each month, except from November 2011 to August 2012, which is at 24:00 on the date 24th or 25th.																																								
Frequency of monitoring/recording	Continuously measured, daily recorded and monthly aggregated																																								
Value monitored	616,762.860																																								
Monitoring equipment	The information of the electricity meters is shown below: <table border="1"> <tr> <th>Location</th><th>M4 installed on line 1</th><th>M5 installed on line 2</th><th>M6 installed on line 3</th></tr> <tr> <td>Type</td><td>DTSD341</td><td>DTSD341</td><td>DTSD341</td></tr> <tr> <td>Accuracy</td><td>0.2S</td><td>0.2S</td><td>0.2S</td></tr> <tr> <td>Serial No.</td><td>09090167290 077</td><td>09090167290 078</td><td>09090167290 079</td></tr> <tr> <td>Calibration frequency</td><td>Yearly</td><td>Yearly</td><td>Yearly</td></tr> <tr> <td>Calibration entity</td><td colspan="3">Yantai Power Supply Company Energy Metering Center/State Grid Shandong Electric Power Metering Center</td></tr> <tr> <td>Date of calibration</td><td>01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016</td><td>01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016</td><td>01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016</td></tr> <tr> <td rowspan="3">Validation</td><td>Yes</td><td>Yes</td><td>Yes</td></tr> <tr> <td colspan="3">From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center</td></tr> <tr> <td colspan="3">From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center</td></tr> </table>			Location	M4 installed on line 1	M5 installed on line 2	M6 installed on line 3	Type	DTSD341	DTSD341	DTSD341	Accuracy	0.2S	0.2S	0.2S	Serial No.	09090167290 077	09090167290 078	09090167290 079	Calibration frequency	Yearly	Yearly	Yearly	Calibration entity	Yantai Power Supply Company Energy Metering Center/State Grid Shandong Electric Power Metering Center			Date of calibration	01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	Validation	Yes	Yes	Yes	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center			From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center		
Location	M4 installed on line 1	M5 installed on line 2	M6 installed on line 3																																						
Type	DTSD341	DTSD341	DTSD341																																						
Accuracy	0.2S	0.2S	0.2S																																						
Serial No.	09090167290 077	09090167290 078	09090167290 079																																						
Calibration frequency	Yearly	Yearly	Yearly																																						
Calibration entity	Yantai Power Supply Company Energy Metering Center/State Grid Shandong Electric Power Metering Center																																								
Date of calibration	01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016																																						
Validation	Yes	Yes	Yes																																						
	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center																																								
	From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center																																								
QA/QC procedures to be applied	The accuracy of meters is 0.2S, which is consistent with the approved revised monitoring plan. Meters were calibrated annually as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000 & DL/T448-2016). Data will be archived for 2 years following the end of the final crediting period.																																								
Purpose of the data	Calculation of baseline emissions																																								
Calculation method	/																																								
Comments	/																																								
Data / Parameter	EG _{import,project}																																								

Data unit	MWh																																						
Description	Electricity imported from the grid by the project																																						
Source of data	Project activity site with electricity meters M4, M5, and M6 installed on the collection line 1, 2, and 3.																																						
Description of measurement methods and procedures to be applied	Electricity imported from the grid by the project is monitored through the bi-directional electricity meters (M4, M5 and M6). The meters (M4, M5 and M6) are installed on the collection lines 1, 2 and 3 of the project activity. The accuracy of above meters is 0.2S. The records of the meters are based on continuous measurement and monthly recorded by the project owner. The cut-off time of recording of all meters is at 24:00 on the last day of each month, except from November 2011 to August 2012, which is at 24:00 on the date 24th or 25th.																																						
Frequency of monitoring/recording	Continuously measured, daily recorded and monthly aggregated																																						
Value monitored	1,623.872																																						
Monitoring equipment	The information of the electricity meters is shown below: <table><tr><td>Location</td><td>M4 installed on line 1</td><td>M5 installed on line 2</td><td>M6 installed on line 3</td></tr><tr><td>Type</td><td>DTSD341</td><td>DTSD341</td><td>DTSD341</td></tr><tr><td>Accuracy</td><td>0.2S</td><td>0.2S</td><td>0.2S</td></tr><tr><td>Serial No.</td><td>09090167290077</td><td>09090167290078</td><td>09090167290079</td></tr><tr><td>Calibration frequency</td><td>Yearly</td><td>Yearly</td><td>Yearly</td></tr><tr><td>Calibration entity</td><td colspan="3">Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center</td></tr><tr><td>Date of calibration</td><td>01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016</td><td>01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016</td><td>01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016</td></tr><tr><td rowspan="3">Validation</td><td>Yes</td><td>Yes</td><td>Yes</td></tr><tr><td colspan="3">From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center</td></tr><tr><td colspan="3">From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center</td></tr></table>	Location	M4 installed on line 1	M5 installed on line 2	M6 installed on line 3	Type	DTSD341	DTSD341	DTSD341	Accuracy	0.2S	0.2S	0.2S	Serial No.	09090167290077	09090167290078	09090167290079	Calibration frequency	Yearly	Yearly	Yearly	Calibration entity	Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center			Date of calibration	01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	Validation	Yes	Yes	Yes	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center			From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center		
Location	M4 installed on line 1	M5 installed on line 2	M6 installed on line 3																																				
Type	DTSD341	DTSD341	DTSD341																																				
Accuracy	0.2S	0.2S	0.2S																																				
Serial No.	09090167290077	09090167290078	09090167290079																																				
Calibration frequency	Yearly	Yearly	Yearly																																				
Calibration entity	Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center																																						
Date of calibration	01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016	01/09/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016																																				
Validation	Yes	Yes	Yes																																				
	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center																																						
	From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center																																						
QA/QC procedures to be applied	The accuracy of meters is 0.2S, which is consistent with the approved revised monitoring plan. Meters were calibrated annually as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000 & DL/T448-2016). Data will be archived for 2 years following the end of the final crediting period.																																						
Purpose of the data	Calculation of baseline emissions																																						

Calculation method	/																													
Comments	/																													
Data / Parameter	$EG_{\text{export,total}}$																													
Data unit	MWh																													
Description	Electricity supplied by the Project and Project B to the grid																													
Source of data	Electricity meters M1 installed at Tangqiu Substation.																													
Description of measurement methods and procedures to be applied	It is continuously measured by the bi-directional meter. M1 and M2 are installed at Tangqiu 220kV Substation. The accuracy of the main meter M1 is 0.2S, and the backup meter M2 is 0.5S. The cut-off time of recording of all meters is at 24:00 on the last day of each month, except from November 2011 to August 2012, which is at 24:00 on the date 24th or 25th.																													
Frequency of monitoring/recording	Continuously measured, daily recorded and monthly aggregated																													
Value monitored	1,187,369.330																													
Monitoring equipment	The information of the electricity meters is shown below: <table border="1"> <tr> <th>Location</th><th>M1 (main meter) installed on 220kV line of Tangqiu Substation</th><th>M2(backup meter) installed on 220kV line of Tangqiu Substation</th></tr> <tr> <td>Type</td><td>ZMQ202C</td><td>DTSD341</td></tr> <tr> <td>Accuracy</td><td>0.2S</td><td>0.5S</td></tr> <tr> <td>Serial No.</td><td>94089552</td><td>20080778020034</td></tr> <tr> <td>Calibration frequency</td><td>Yearly</td><td>Yearly</td></tr> <tr> <td>Calibration entity</td><td colspan="2">Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center</td></tr> <tr> <td>Date of calibration</td><td>09/11/2011; 17/09/2012; 12/09/2013; 14/03/2014; 10/03/2015; 08/03/2016</td><td>09/11/2011; 17/09/2012; 12/09/2013; 14/03/2014; 10/03/2015; 08/03/2016</td></tr> <tr> <td rowspan="3">Validation</td><td>Yes</td><td>Yes</td></tr> <tr> <td colspan="2">From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center</td></tr> <tr> <td colspan="2">From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center</td></tr> </table>		Location	M1 (main meter) installed on 220kV line of Tangqiu Substation	M2(backup meter) installed on 220kV line of Tangqiu Substation	Type	ZMQ202C	DTSD341	Accuracy	0.2S	0.5S	Serial No.	94089552	20080778020034	Calibration frequency	Yearly	Yearly	Calibration entity	Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center		Date of calibration	09/11/2011; 17/09/2012; 12/09/2013; 14/03/2014; 10/03/2015; 08/03/2016	09/11/2011; 17/09/2012; 12/09/2013; 14/03/2014; 10/03/2015; 08/03/2016	Validation	Yes	Yes	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center		From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center	
Location	M1 (main meter) installed on 220kV line of Tangqiu Substation	M2(backup meter) installed on 220kV line of Tangqiu Substation																												
Type	ZMQ202C	DTSD341																												
Accuracy	0.2S	0.5S																												
Serial No.	94089552	20080778020034																												
Calibration frequency	Yearly	Yearly																												
Calibration entity	Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center																													
Date of calibration	09/11/2011; 17/09/2012; 12/09/2013; 14/03/2014; 10/03/2015; 08/03/2016	09/11/2011; 17/09/2012; 12/09/2013; 14/03/2014; 10/03/2015; 08/03/2016																												
Validation	Yes	Yes																												
	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center																													
	From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center																													
QA/QC procedures to	The accuracy of meters M1 and M2 is 0.2S and 0.5 respectively, which is consistent with approved revised monitoring plan. Meters																													

be applied	were calibrated annually as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000 & DL/T448-2016). Meter readings of $EG_{\text{export,total}}$ are double checked with electricity sales receipts. Data will be archived for 2 years following the end of the final crediting period.
Purpose of the data	Calculation of baseline emissions
Calculation method	/
Comments	/

Data / Parameter	$EG_{\text{import,total}}$																											
Data unit	MWh																											
Description	Electricity imported from the grid to the Project and Project B																											
Source of data	Electricity meter M1 installed at Tangqiu Substation.																											
Description of measurement methods and procedures to be applied	It is continuously measured by the bi-directional meters M1 and M2 are installed at Tangqiu 220kV Substation. The accuracy of the main meter M1 is 0.2S, and the backup meter M2 is 0.5S. The cut-off time of recording of all meters is at 24:00 on the last day of each month, except from November 2011 to August 2012, which is at 24:00 on the 24th or 25th																											
Frequency of monitoring/recording	Continuously measured and daily recorded and monthly aggregated																											
Value monitored	3,764.640																											
Monitoring equipment	The information of the electricity meters is shown below: <table border="1"> <tr> <th>Location</th><th>M1 (main meter) installed on 220kV line of Tangqiu Substation</th><th>M2(backup meter) installed on 220kV line of Tangqiu Substation</th></tr> <tr> <td>Type</td><td>ZMQ202C</td><td>DTSD341</td></tr> <tr> <td>Accuracy</td><td>0.2S</td><td>0.5S</td></tr> <tr> <td>Serial No.</td><td>94089552</td><td>20080778020034</td></tr> <tr> <td>Calibration frequency</td><td>Yearly</td><td>Yearly</td></tr> <tr> <td>Calibration entity</td><td colspan="2">Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center</td></tr> <tr> <td>Date of calibration</td><td>09/11/2011; 17/09/2012; 12/09/2013; 14/03/2014; 10/03/2015; 08/03/2016</td><td>09/11/2011; 17/09/2012; 12/09/2013; 14/03/2014; 10/03/2015; 08/03/2016</td></tr> <tr> <td rowspan="2">Validation</td><td>Yes</td><td>Yes</td></tr> <tr> <td colspan="2">From 2011 to 2014, it was verified by</td></tr> </table>		Location	M1 (main meter) installed on 220kV line of Tangqiu Substation	M2(backup meter) installed on 220kV line of Tangqiu Substation	Type	ZMQ202C	DTSD341	Accuracy	0.2S	0.5S	Serial No.	94089552	20080778020034	Calibration frequency	Yearly	Yearly	Calibration entity	Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center		Date of calibration	09/11/2011; 17/09/2012; 12/09/2013; 14/03/2014; 10/03/2015; 08/03/2016	09/11/2011; 17/09/2012; 12/09/2013; 14/03/2014; 10/03/2015; 08/03/2016	Validation	Yes	Yes	From 2011 to 2014, it was verified by	
Location	M1 (main meter) installed on 220kV line of Tangqiu Substation	M2(backup meter) installed on 220kV line of Tangqiu Substation																										
Type	ZMQ202C	DTSD341																										
Accuracy	0.2S	0.5S																										
Serial No.	94089552	20080778020034																										
Calibration frequency	Yearly	Yearly																										
Calibration entity	Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center																											
Date of calibration	09/11/2011; 17/09/2012; 12/09/2013; 14/03/2014; 10/03/2015; 08/03/2016	09/11/2011; 17/09/2012; 12/09/2013; 14/03/2014; 10/03/2015; 08/03/2016																										
Validation	Yes	Yes																										
	From 2011 to 2014, it was verified by																											

	Yantai Power Supply Company Energy Metering Center
	From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center
QA/QC procedures to be applied	The accuracy of meters M1 and M2 is 0.2S and 0.5S respectively, which is consistent with approved revised monitoring plan. Meters were calibrated annually as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000 & DL/T448-2016). Meter readings of $EG_{import, total}$ are double checked with electricity sales receipts. Data will be archived for 2 years following the end of the final crediting period.
Purpose of the data	Calculation of baseline emissions
Calculation method	/
Comments	/

Data / Parameter	EG _{export,other,y}																					
Data unit	MWh																					
Description	Electricity supplied by the Project B to the grid																					
Source of data	Electricity meters M7and M8 installed on the collection line 4 and 5 at Project site.																					
Description of measurement methods and procedures to be applied	Electricity supplied to the grid by Project B is monitored through the bi-directional electricity meters (main meters M7 and M8, and backup meters M9 and M10) installed on the collection lines 4 and 5. The accuracy of the main meters (M7 and M8) and backup meters (M9 and M10) is all 0.5 s. The cut-off time of recording of all meters is at 24:00 on the last day of each month, except from November 2011 to August 2012, which is at 24:00 on the date 24th or 25th.																					
Frequency of monitoring/recording	Continuously measured and daily recorded and monthly aggregated																					
Value monitored	579,392.800																					
Monitoring equipment	The information of the electricity meters is shown below: <table><tr><td>Location</td><td>M7 (main meter) installed on line 4</td><td>M8(Main meter) installed on line 5</td></tr><tr><td>Type</td><td>DTSD341</td><td>DTSD341</td></tr><tr><td>Accuracy</td><td>0.5S</td><td>0.5S</td></tr><tr><td>Serial No.</td><td>200908326835</td><td>200908326832</td></tr><tr><td>Calibration frequency</td><td>Yearly</td><td>Yearly</td></tr><tr><td>Calibration entity</td><td colspan="2">Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center</td></tr><tr><td>Date of calibration</td><td>09/11/2011; 11/09/2012;</td><td>09/11/2011; 11/09/2012;</td></tr></table>	Location	M7 (main meter) installed on line 4	M8(Main meter) installed on line 5	Type	DTSD341	DTSD341	Accuracy	0.5S	0.5S	Serial No.	200908326835	200908326832	Calibration frequency	Yearly	Yearly	Calibration entity	Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center		Date of calibration	09/11/2011; 11/09/2012;	09/11/2011; 11/09/2012;
Location	M7 (main meter) installed on line 4	M8(Main meter) installed on line 5																				
Type	DTSD341	DTSD341																				
Accuracy	0.5S	0.5S																				
Serial No.	200908326835	200908326832																				
Calibration frequency	Yearly	Yearly																				
Calibration entity	Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center																					
Date of calibration	09/11/2011; 11/09/2012;	09/11/2011; 11/09/2012;																				

		10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016,	10/09/2013; 11/03/2014; 10/03/2015; 08/03/2016
	Validation	Yes	Yes
		From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center	
		From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center	
	Location	M9 (Backup meter) installed on line 4	M10 (Backup meter) installed on line 5
	Type	DTSD341	DTSD341
	Accuracy	0.5S	0.5S
	Serial No.	200908326874	200908326873
	Calibration frequency	Yearly	Yearly
Calibration entity	Yantai Power Supply Company Energy Metering Center / State Grid Shandong Electric Power Metering Center		
Date of calibration	09/11/2011; 11/09/2012; 10/09/2013; 11/03/2014; 10/03/2015; 08/03/2015	09/11/2011, 11/09/2012, 10/09/2013, 11/03/2014, 10/03/2015; 08/03/2016	
Validation	Yes	Yes	
	From 2011 to 2014, it was verified by Yantai Power Supply Company Energy Metering Center		
	From 2015 to 2016, it was verified by State Grid Shandong Electric Power Metering Center		
QA/QC procedures to be applied	The accuracy of meters M7, M8, M9 and M10 is 0.5S, which is consistent with the approved revised monitoring plan. Meters were calibrated annually as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000 & DL/T448-2016). Data will be archived for 2 years following the end of the final crediting period.		
Purpose of the data	Calculation of baseline emissions		
Calculation method	/		
Comments	/		

Data / Parameter	EG_y
Data unit	MWh
Description	Net electricity supplied to the grid by the project activity
Source of data	Calculation by $EG_{export,total} - EG_{import,total} - EG_{export,other}$ and cross-check with $EG_{export,project} - EG_{import,project}$.
Description of measurement methods and procedures to be applied	The net electricity supplied to the grid by the project is calculated by $EG_{export,total} - EG_{import,total} - EG_{export,other}$ and cross-check with $EG_{export,project} - EG_{import,project}$. The conservative values were used for the calculation of emission reductions.
Frequency of monitoring/recording	Continuously measured, daily recorded and monthly aggregated
Value monitored	598,594.662
Monitoring equipment	The information of the monitoring equipments of the related monitoring parameters have been listed on the above tables.
QA/QC procedures to be applied	All the meters' accuracy are consistent with approved revised monitoring plan. Meters were calibrated annually as per the requirement of Technical administrative code of electric energy metering (DL/T448-2000 & DL/T448-2016). Net electricity supplied to the grid are calculated by " $EG_{export,total} - EG_{import,total} - EG_{export,other}$ " has been cross-checked with " $EG_{export,project} - EG_{import,project}$ " ($EG_{export,total}$, $EG_{import,total}$ and $EG_{export,other}$ were cross-checked with the sales receipts issued by grid company). Data will be archived for 2 years following the end of the final crediting period.
Purpose of the data	Calculation of baseline emissions
Calculation method	/
Comments	/

4.3 Monitoring Plan

1. Monitoring system organization

China Resources Wind Power (Yantai) Co., Ltd., the owner of the project, use this document as guideline in monitoring of the project emission reduction performance and adhere to the guidelines set out in this monitoring plan to ensure that the monitoring is credible, transparent and conservative.

The responsibilities of the project staff are as follow:

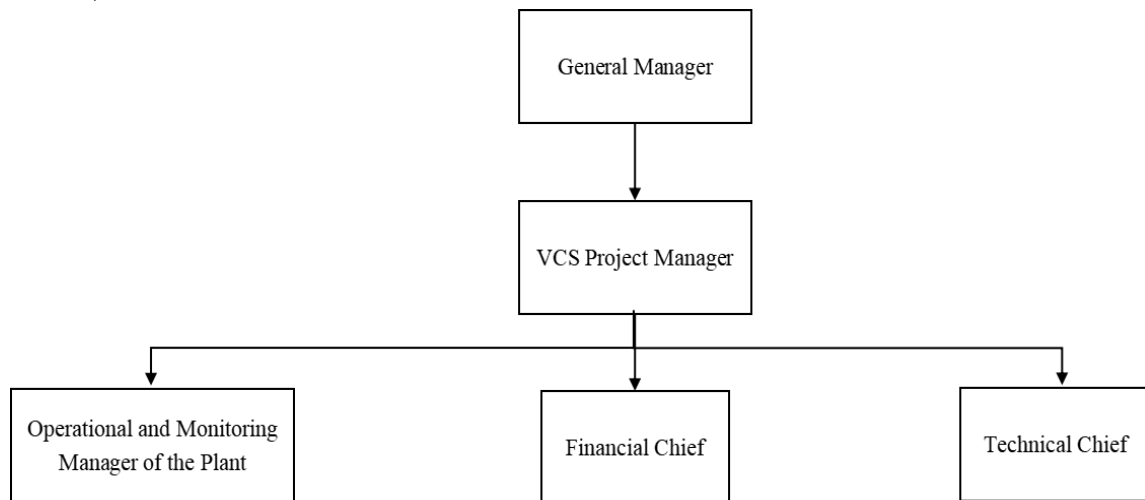
General Manager: To be responsible for supervising the whole monitoring procedure.

VCS Project Manager: To be responsible for data management and compiling monitoring report.

Operational and monitoring manager: To be responsible for collecting data and do internal audit.

Financial chief: To be responsible for collection of sales receipts.

Technical chief: To be responsible for preparing operational reports of the project activity, recording the daily operation of the wind farm, including operating periods, equipment defects, etc.



2. Data collection procedures

The total quantity of electricity supplied to the grid and imported from the grid by the project activity and Project B ($EG_{export,total}$ and $EG_{import,total}$) are monitored through the bi-directional meters (main meter M1 and backup meter M2) installed at Tangqiu 220kV Substation. The accuracy of the main meter M1 is 0.2S and the accuracy of the backup meter M2 is 0.5S. The records of the meters are based on continuous measurement, daily recorded and monthly aggregated by the project owner. Monthly total quantity of the electricity supplied to and imported from the grid by the project activity and Project B is approved and signed off by the Manager before it is accepted and stored. The readings of the meter M1 are monitored and recorded on continuous basis and aggregated on monthly basis by the grid company. According to the PPA signed between the project owner and the grid company, the cut-off time of $EG_{export,total}$ and $EG_{import,total}$ is at 24:00 on the last day of each month, except from November 2011 to August 2012, which is at 24:00 on the date 24th or 25th. And the sales receipts were issued based on the amount confirmed by both sides and were used to cross-check the meters' readings.

Electricity supplied to the grid by Project B ($EG_{export,other}$) is monitored through the bi-directional electricity meters (main meters M7 and M8, and backup meters M9 and M10) installed on the collection lines 4 and 5. The accuracy of the main meters (M7 and M8) and

backup meters (M9 and M10) is 0.5S. The records of the meters are based on continuous measurement, daily recorded and monthly aggregated by the project owner. Monthly electricity supplied to the grid by Project B is approved and signed off by the Manager before it is accepted and stored. The readings of the meters are also monitored and recorded on continuous basis and aggregated on monthly basis by the grid company. According to the PPA signed between the project owner and the grid company, sales receipts are issued according to meter readings of meters (M7 and M8) installed on the collection lines 4 and 5 to cross-check the reading records of $EG_{\text{export,other}}$.

Electricity supplied to the grid and imported from the grid by the project activity ($EG_{\text{export,project}}$ and $EG_{\text{import,project}}$) is monitored through the bi-directional electricity meters (M4, M5 and M6). The meters (M4, M5 and M6) are installed on the collection lines 1, 2 and 3 of the project activity. The accuracy of above meters is 0.2S. The records of the meters are based on continuous measurement and monthly recorded by the project owner. Monthly electricity supplied to the grid and imported from the grid by the project activity is approved and signed off by the Manager before it is accepted and stored.

The metering equipment is calibrated and checked for accuracy so that the metering equipments have sufficient accuracy within the agreed limits. The metering equipments are calibrated and checked annually by qualified third party for accuracy according to national standard. The records are supplied to the project owner, and maintained by the project owner.

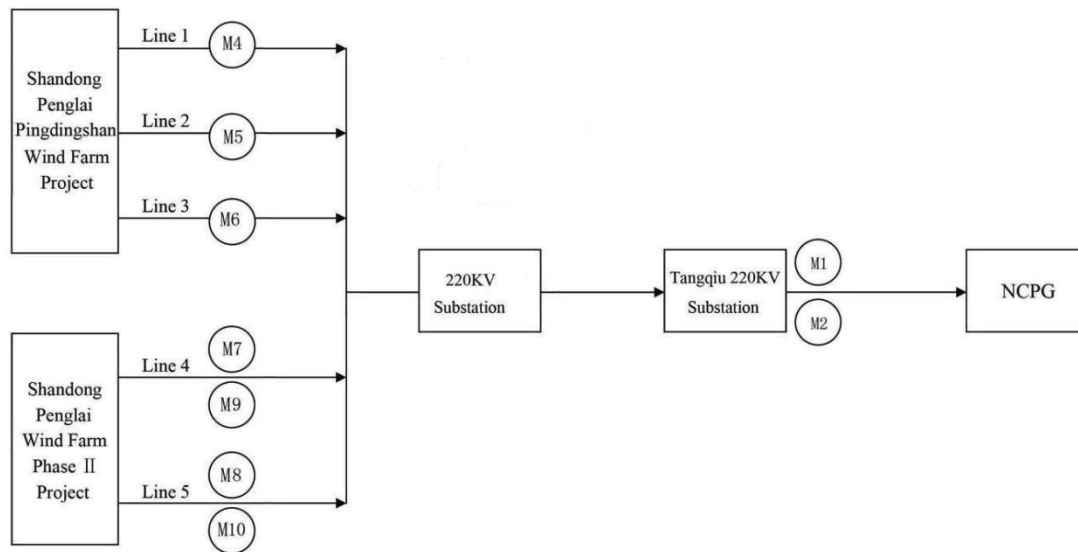
Therefore, the net electricity supplied to the grid by the Project is calculated with the following formula, and double checked with relevant sales receipts:

$$EG_y = EG_{\text{export,total}} - EG_{\text{import,total}} - EG_{\text{export,other}}$$

The above metered net electricity supplied to the grid by the Project is cross-checked with the following metered readings:

$$EG_y = EG_{\text{export,project}} - EG_{\text{import,project}}$$

The below diagram shows location of the electricity meters installed:



3. Quality Assurance and Quality Control Procedure

The workers are trained to be competent and the metering equipments are calibrated and sealed as per the industry practices at regular intervals, with the purpose to provide credible, accurate, transparent and conservative monitoring data and ensure the real, measurable, long-term GHG emission reduction from the Project.

Net electricity supplied to the grid are calculated by “ $EG_{\text{export,total}} - EG_{\text{import,total}} - EG_{\text{export,other}}$ ” has been cross-checked with “ $EG_{\text{export,project}} - EG_{\text{import,project}}$ ”, and $EG_{\text{export,other}}$ were cross-checked with the sales receipts issued by grid company. The metered electricity supplied to the grid by the Project and Project B ($EG_{\text{export,total}}$) is double checked with the sales receipts issued based on meter readings from meter M1 installed at Tangqiu 220kV Substation. To be conservative, the lower value is applied for the calculation.

The metered electricity imported from the grid by the Project ($EG_{\text{import,total}}$) is double checked with the sales receipts issued based on meter readings from meter M1 installed at Tangqiu 220kV Substation, which is the total quantity of electricity imported from the grid by the Project and Project B. To be conservative, the higher value is applied for the calculation of emission reductions of the Project.

The metered electricity supplied to the grid by the Project B ($EG_{\text{export, other}}$) is double checked with the sales receipts issued based on meter readings from meter M7 installed on the collection line 4 and meter 8 installed on the collection line 5 of the project B. To be conservative, the higher value is applied for the calculation.

Monthly net electricity supplied to the grid by the project activity is approved and signed off by the Manager before it is accepted and stored. This audit checks compliance with monitoring procedures in the revised monitoring plan. This internal audit identifies potential improvements to procedures to improve monitoring and reporting in future years.

Emergency Procedure:

In case metering equipment is damaged and no reliable readings can be recorded the project entity will estimate net supply by the Project according to the following procedure:

1. In case the main meters are damaged, the meter readings from the backup meters will be used as record of net power supplied to the grid by the project activity for the days when no record could be recorded.
2. In case both the main meters and backup meters are damaged: The project owner and the grid company will jointly calculate a conservative estimate of power supplied to the grid. A statement will be prepared indicating:
 - the background to the damage to metering equipment;
 - the assumptions used to estimate net supply to the grid for the days for which no record could be recorded;
 - the estimation of power supplied to the grid. The statement will be signed by both a representative of the project entity as well as a representative of the grid company.

If any error is detected, the party owning the meter shall repair, recalibrate or replace the meter giving the other party sufficient notice to allow a representative to attend during any corrective activity.

The project owner will furthermore document all efforts taken to restore normal monitoring procedures.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

According to ACM0002 and the registered VCS PD of the Project, the baseline emission BE_y during the monitoring period results from:

$$BE_y = EG_y \times EF_{\text{grid,CM,y}}$$

Where:

EG_y – Net electricity supplied to the grid by the project activity in year y;

$EF_{\text{grid,CM,y}}$ - Combined margin CO₂ emission factor for the project electricity system in year y (tCO₂e/MWh).

Table 3: Calculation of baseline emission reductions

Monitoring period	EG_y	$EF_{\text{grid,CM,y}}$	BE_y
	Unit: MWh	Unit: tCO ₂ e/MWh	Unit: tCO ₂ e
26/10/2011-31/12/2011	22,531.872	1.0548	23,766
01/01/2012-31/12/2012	118,974.684	1.0548	125,494
01/01/2013-31/12/2013	122,916.848	1.0548	129,652
01/01/2014-31/12/2014	111,861.458	1.0548	117,991
01/01/2015-31/12/2015	109,738.078	1.0548	115,751
01/01/2016-31/12/2016	112,571.722	1.0548	118,740
Total	598,594.662	1.0548	631,397

The monthly data, crosscheck and calculation procedure are shown in appendix 1.

5.2 Project Emissions

According to ACM0002(version 07), the project does not have any GHG emissions.

$$PE_y = 0$$

5.3 Leakage

According to ACM0002 (Version 07), the leakage from the Project is zero.

$L_y=0$

5.4 Net GHG Emission Reductions and Removals

Total baseline emissions: $BE_y = 631,397 \text{ tCO}_2\text{e}$

Total project emissions: $PE_y = 0$

Total leakage: $L_y = 0$

Total emission reductions achieved during this monitoring period from 26/10/2011 to 31/12/2016: $ER_y = BE_y - PE_y - L_y = 631,397 - 0 - 0 = 631,397 \text{ tCO}_2\text{e}$.

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals(tCO ₂ e)
26/10/2011-31/12/2011	23,766	0	0	23,766
01/01/2012-31/12/2012	125,494	0	0	125,494
01/01/2013-31/12/2013	129,652	0	0	129,652
01/01/2014-31/12/2014	117,991	0	0	117,991
01/01/2015-31/12/2015	115,751	0	0	115,751
01/01/2016-31/12/2016	118,740	0	0	118,740
Total	631,397	0	0	631,397

As per the registered PDD, the estimated quantity of annual emission reductions for the project activity is 107,488 tCO₂e. The estimated emission reductions during this monitoring period from 26/10/2011 to 31/12/2016 (1893 days) are 557,465.162 tCO₂e while the actual emission reductions are 631,397 tCO₂e, which is 13% higher than the estimation emission reductions.

This would not affect the additionality of the project. As per to the registered PDD, the IRR is still less than the beach mark of 8% when the net electricity supplied to the grid increase 13%. Only when the electricity generation increase by 16%, the IRR could reach benchmark.

The reasons for electricity and emission reductions increase are as follows: The data source of annual electricity supplied to the grid from the project in the registered PDD is from the Feasibility Study Report (FSR) of the project, which is calculated based on the monitored wind resources data from the project onsite anemometer tower during year 2005 to 2006 and the monitored wind resources data from local Weather Station during year 1986 to 2006.

Comparing the yearly average wind from 2012-2016 and that from 1987- 2006, the wind speed in this period is 65% higher than the data used in the FSR (refer to figure 2 for details). Furthermore, the Opti Speed generators have been used by the project, which allows the

rotor speed to vary within a range of approximately 60 percent in relation to normal rpm. So it's reasonable that the net electricity supplied to the grid by the project during this monitoring period is 13% higher than the estimated data because of the rich wind resources, Opti Speed generators and the better management.

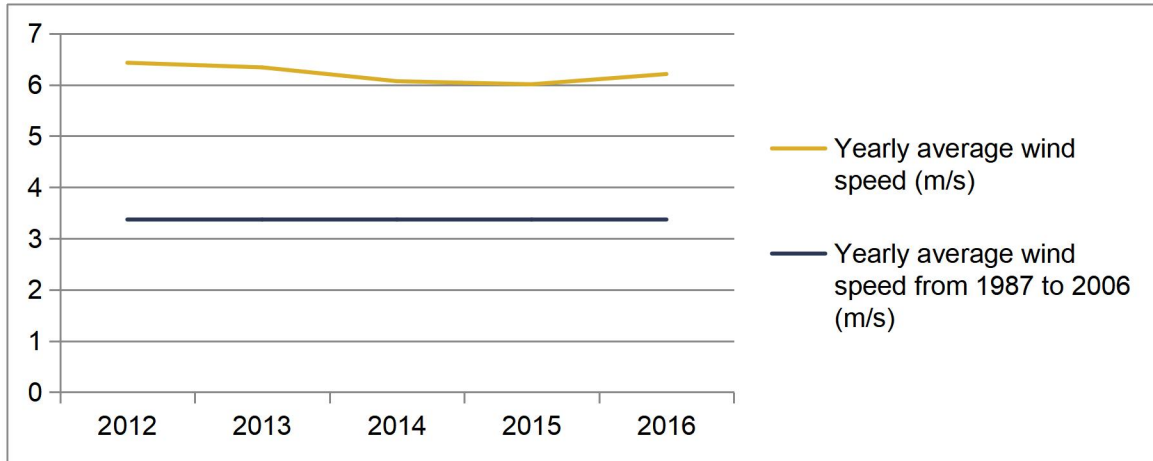


Figure 2. Comparison of wind speed from 2012-2016 with 1987-2006

APPENDIX 1: < MONTHLY AND YEARLY ELECTRICITY SUPPLY >

Table 4: Electricity supplied by the Project and Project B to the grid and Electricity imported from the grid to the Project and Project B

Monitoring period	EG _{export,total}			EG _{import,total}		
	Data based on meter readings M1	Data based on sales receipt	Conservative value	Data based on meter readings M1	Data based on sales receipt	Conservative value
	MWh	MWh	MWh	MWh	MWh	MWh
	A	B	C	D	E	F
26/10/2011-24/11/2011	20,193.200	20,193.200	20,193.200	60.720	60.720	60.720
25/11/2011-25/12/2011	20,971.160	20,971.160	20,971.160	52.800	52.800	52.800
26/12/2011-31/12/2011	3,405.600	3,405.600	3,405.600	0.000	0.000	0.000
Subtotal 2011	-	-	44,569.960	-	-	113.520
01/01/2012-25/01/2012	9,643.920	9,643.920	9,643.920	0.000	0.000	0.000
25/01/2012-24/02/2012	18,831.120	18,831.120	18,831.120	150.480	150.480	150.480
25/02/2012-25/03/2012	18,987.800	18,987.800	18,987.800	47.520	47.520	47.520
26/03/2012-24/04/2012	32,244.960	32,244.960	32,244.960	26.400	26.400	26.400
25/04/2012-25/05/2012	19,272.000	19,272.000	19,272.000	71.280	71.280	71.280
24/05/2012-30/06/2012	25,098.480	25,098.480	25,098.480	63.360	63.360	63.360
01/07/2012-31/07/2012	14,688.960	14,688.960	14,688.960	84.480	84.480	84.480
01/08/2012-23/08/2012	6,473.280	6,473.280	6,473.280	105.600	105.600	105.600
24/08/2012-30/09/2012	18,004.800	18,004.800	18,004.800	92.400	92.400	92.400
01/10/2012-31/10/2012	19,715.520	19,715.520	19,715.520	52.800	52.800	52.800
01/11/2012-30/11/2012	26,967.600	26,967.600	26,967.600	34.320	34.320	34.320
01/12/2012-31/12/2012	24,546.720	24,546.720	24,546.720	39.600	39.600	39.600
Subtotal 2012	-	-	234,475.160	-	-	768.240

01/01/2013-31/01/2013	18,308.400	18,308.400	18,308.400	42.240	42.240	42.240
01/02/2013-28/02/2013	12,608.640	12,608.640	12,608.640	147.840	147.840	147.840
01/03/2013-31/03/2013	31,658.880	31,658.880	31,658.880	18.480	18.480	18.480
01/04/2013-30/04/2013	30,101.280	30,101.280	30,101.280	29.040	29.040	29.040
01/05/2013-31/05/2013	23,617.440	23,617.440	23,617.440	34.320	34.320	34.320
01/06/2013-30/06/2013	19,430.400	19,430.400	19,430.400	58.080	58.080	58.080
01/07/2013-31/07/2013	13,453.440	13,453.440	13,453.440	81.840	81.840	81.840
01/08/2013-31/08/2013	14,343.120	14,343.120	14,343.120	76.560	76.560	76.560
01/09/2013-30/09/2013	11,993.520	11,993.520	11,993.520	100.320	100.320	100.320
01/10/2013-31/10/2013	20,787.360	20,787.360	20,787.360	42.240	42.240	42.240
01/11/2013-30/11/2013	22,672.320	22,672.320	22,672.320	58.080	58.080	58.080
01/12/2013-31/12/2013	24,192.960	24,192.960	24,192.960	18.480	18.480	18.480
Subtotal 2013	-	-	243,167.760	-	-	707.520
01/01/2014-31/01/2014	20,132.640	20,132.640	20,132.640	23.760	23.760	23.760
01/02/2014-28/02/2014	21,209.760	21,209.760	21,209.760	52.800	52.800	52.800
01/03/2014-31/03/2014	21,492.240	21,492.240	21,492.240	34.320	34.320	34.320
01/04/2014-30/04/2014	18,973.680	18,973.680	18,973.680	58.080	58.080	58.080
01/05/2014-31/05/2014	29,924.400	29,924.400	29,924.400	29.040	29.040	29.040
01/06/2014-30/06/2014	11,528.880	11,528.880	11,528.880	60.720	60.720	60.720
01/07/2014-31/07/2014	15,253.920	15,253.920	15,253.920	81.840	81.840	81.840
01/08/2014-31/08/2014	6,378.240	6,378.240	6,378.240	132.000	132.000	132.000
01/09/2014-30/09/2014	9,065.760	9,065.760	9,065.760	81.840	81.840	81.840
01/10/2014-31/10/2014	22,297.440	22,297.440	22,297.440	52.800	52.800	52.800
01/11/2014-30/11/2014	15,768.720	15,768.720	15,768.720	39.600	39.600	39.600
01/12/2014-31/12/2014	29,480.880	29,480.880	29,480.880	44.880	44.880	44.880
Subtotal 2014	-	-	221,506.560	-	-	691.680

01/01/2015-31/01/2015	23,810.160	23,810.160	23,810.160	18.480	18.480	18.480
01/02/2015-28/02/2015	16,101.360	16,101.360	16,101.360	60.720	60.720	60.720
01/03/2015-31/03/2015	25,130.160	25,130.160	25,130.160	42.240	42.240	42.240
01/04/2015-30/04/2015	28,511.470	28,511.470	28,511.470	39.600	39.600	39.600
01/05/2015-31/05/2015	22,968.000	22,968.000	22,968.000	54.120	54.120	54.120
01/06/2015-30/06/2015	16,518.220	16,518.220	16,518.220	80.784	80.784	80.784
01/07/2015-31/07/2015	11,064.240	11,064.240	11,064.240	100.056	100.056	100.056
01/08/2015-31/08/2015	5,733.290	5,733.290	5,733.290	120.120	120.120	120.120
01/09/2015-30/09/2015	7,661.280	7,661.280	7,661.280	135.696	135.696	135.696
01/10/2015-31/10/2015	18,996.910	18,996.910	18,996.910	69.168	69.168	69.168
01/11/2015-30/11/2015	22,885.900	22,885.900	22,885.900	56.496	56.496	56.496
01/12/2015-31/12/2015	21,495.140	21,495.140	21,495.140	53.592	53.592	53.592
Subtotal 2015	-	-	220,876.130	-	-	831.072
01/01/2016-31/01/2016	22,600.780	22,600.780	22,600.780	68.904	68.904	68.904
01/02/2016-28/02/2016	15,923.690	15,923.690	15,923.690	43.560	43.560	43.560
01/03/2016-31/03/2016	22,322.520	22,322.520	22,322.520	34.320	34.320	34.320
01/04/2016-30/04/2016	24,050.660	24,050.660	24,050.660	27.720	27.720	27.720
01/05/2016-31/05/2016	18,280.680	18,280.680	18,280.680	27.456	27.456	27.456
01/06/2016-30/06/2016	19,477.390	19,477.390	19,477.390	77.352	77.352	77.352
01/07/2016-31/07/2016	14,986.750	14,986.750	14,986.750	77.352	77.352	77.352
01/08/2016-31/08/2016	11,792.880	11,792.880	11,792.880	78.936	78.936	78.936
01/09/2016-30/09/2016	14,611.080	14,611.080	14,611.080	101.904	101.904	101.904
01/10/2016-31/10/2016	15,267.120	15,267.120	15,267.120	50.424	50.424	50.424
01/11/2016-30/11/2016	21,792.940	21,792.940	21,792.940	44.088	44.088	44.088
01/12/2016-31/12/2016	21,667.270	21,667.270	21,667.270	20.592	20.592	20.592
Subtotal 2016	-	-	222,773.760	-	-	652.608

Total	-	-	1,187,369.330	-	-	3,764.640
--------------	---	---	----------------------	---	---	------------------

Table 5: Electricity supplied by the Project B to the grid and Net electricity supplied to the grid by the project activity

Monitoring period	EG _{export,other,y}					EG _y
	Data based on meter readings			Data based on sales receipt	Conservative value	Calculated by EG _{export,total} - EG _{import,total} - EG _{export,other}
	Meter readings based on M7	Meter readings based on M8	Subtotal			
	MWh	MWh	MWh	MWh	MWh	MWh
	G	H	I=G+H	J	K	L=C-F-K
26/10/2011-24/11/2011	6,205.850	4,519.340	10,725.190	10,725.190	10,725.190	9,407.290
25/11/2011-25/12/2011	5,183.500	3,701.180	8,884.680	8,884.680	8,884.680	12,033.680
26/12/2011-31/12/2011	940.450	626.500	1,566.950	1,566.950	1,566.950	1,838.650
Subtotal 2011	-	-	-	-	21,176.820	23,279.620
01/01/2012-25/01/2012	2,731.400	1,868.720	4,600.120	4,600.120	4,600.120	5,043.800
26/01/2012-24/02/2012	5,414.150	3,664.360	9,078.510	9,078.510	9,078.510	9,602.130
25/02/2012-25/03/2012	5,057.150	3,866.240	8,923.390	8,923.390	8,923.390	10,016.890
26/03/2012-24/04/2012	9,372.930	7,111.300	16,484.230	16,484.230	16,484.230	15,734.330
25/04/2012-25/05/2012	5,056.100	4,210.220	9,266.320	9,266.320	9,266.320	9,934.400
26/05/2012-30/06/2012	6,950.300	5,114.900	12,065.200	12,065.200	12,065.200	12,969.920
01/07/2012-31/07/2012	4,236.400	2,774.520	7,010.920	7,010.920	7,010.920	7,593.560
01/08/2012-23/08/2012	1,924.300	1,245.160	3,169.460	3,169.460	3,169.460	3,198.220
24/08/2012-30/09/2012	5,205.200	3,252.760	8,457.960	8,457.960	8,457.960	9,454.440
01/10/2012-31/10/2012	5,520.900	3,970.120	9,491.020	9,491.020	9,491.020	10,171.700
01/11/2012-30/11/2012	7,530.250	5,731.600	13,261.850	13,261.850	13,261.850	13,671.430
01/12/2012-31/12/2012	6,933.150	4,899.160	11,832.310	11,832.310	11,832.310	12,674.810

Subtotal 2012	-	-	-	-	113,641.290	120,065.630
01/01/2013-31/01/2013	5,216.750	3,450.720	8,667.470	8,667.470	8,667.470	9,598.690
01/02/2013-28/02/2013	3,555.650	2,603.160	6,158.810	6,158.810	6,158.810	6,301.990
01/03/2013-31/03/2013	8,317.400	7,288.680	15,606.080	15,606.080	15,606.080	16,034.320
01/04/2013-30/04/2013	8,352.750	6,558.160	14,910.910	14,910.910	14,910.910	15,161.330
01/05/2013-31/05/2013	6,647.900	4,994.080	11,641.980	11,641.980	11,641.980	11,941.140
01/06/2013-30/06/2013	5,439.350	3,862.880	9,302.230	9,302.230	9,302.230	10,070.090
01/07/2013-31/07/2013	3,877.300	2,582.160	6,459.460	6,459.460	6,459.460	6,912.140
01/08/2013-31/08/2013	4,000.150	2,795.240	6,795.390	6,795.390	6,795.390	7,471.170
01/09/2013-30/09/2013	3,384.150	2,388.680	5,772.830	5,772.830	5,772.830	6,120.370
01/10/2013-31/10/2013	5,784.450	4,340.840	10,125.290	10,125.290	10,125.290	10,619.830
01/11/2013-30/11/2013	6,249.250	4,678.240	10,927.490	10,927.490	10,927.490	11,686.750
01/12/2013-31/12/2013	6,807.150	4,745.440	11,552.590	11,552.590	11,552.590	12,621.890
Subtotal 2013	-	-	-	-	117,920.530	124,539.710
01/01/2014-31/01/2014	5,553.450	4,263.000	9,816.450	9,816.450	9,816.450	10,292.430
01/02/2014-28/02/2014	5,750.500	4,889.080	10,639.580	10,639.580	10,639.580	10,517.380
01/03/2014-31/03/2014	6,045.550	4,866.680	10,912.230	10,912.230	10,912.230	10,545.690
01/04/2014-30/04/2014	5,184.200	3,971.240	9,155.440	9,155.440	9,155.440	9,760.160
01/05/2014-31/05/2014	7,998.900	6,256.040	14,254.940	14,254.940	14,254.940	15,640.420
01/06/2014-30/06/2014	3,493.000	2,579.920	6,072.920	6,072.920	6,072.920	5,395.240
01/07/2014-31/07/2014	4,049.150	3,115.000	7,164.150	7,164.150	7,164.150	8,007.930
01/08/2014-31/08/2014	1,942.500	1,285.480	3,227.980	3,227.980	3,227.980	3,018.260
01/09/2014-30/09/2014	2,501.450	1,855.560	4,357.010	4,357.010	4,357.010	4,626.910
01/10/2014-31/10/2014	5,845.700	4,852.680	10,698.380	10,698.380	10,698.380	11,546.260
01/11/2014-30/11/2014	4,472.650	3,355.240	7,827.890	7,827.890	7,827.890	7,901.230
01/12/2014-31/12/2014	7,835.450	6,312.040	14,147.490	14,147.490	14,147.490	15,288.510

Subtotal 2014	-	-	-	-	108,274.460	112,540.420
01/01/2015-31/01/2015	6,725.600	5,237.400	11,963.000	11,963.000	11,963.000	11,828.680
01/02/2015-28/02/2015	4,585.000	3,462.480	8,047.480	8,047.480	8,047.480	7,993.160
01/03/2015-31/03/2015	6,470.100	5,278.280	11,748.380	11,748.380	11,748.380	13,339.540
01/04/2015-30/04/2015	8,271.200	6,502.720	14,773.920	14,773.920	14,773.920	13,697.950
01/05/2015-31/05/2015	6,485.150	4,981.200	11,466.350	11,466.350	11,466.350	11,447.530
01/06/2015-30/06/2015	4,404.750	3,518.480	7,923.230	7,923.230	7,923.230	8,514.206
01/07/2015-31/07/2015	3,289.650	2,311.120	5,600.770	5,600.770	5,600.770	5,363.414
01/08/2015-31/08/2015	1,612.800	1,270.640	2,883.440	2,883.440	2,883.440	2,729.730
01/09/2015-30/09/2015	2,129.750	1,258.880	3,388.630	3,388.630	3,388.630	4,136.954
01/10/2015-31/10/2015	5,756.800	4,335.520	10,092.320	10,092.320	10,092.320	8,835.422
01/11/2015-30/11/2015	6,378.050	4,868.920	11,246.970	11,246.970	11,246.970	11,582.434
01/12/2015-31/12/2015	5,950.350	4,525.080	10,475.430	10,475.430	10,475.430	10,966.118
Subtotal 2015	-	-	-	-	109,609.920	110,435.138
01/01/2016-31/01/2016	6,329.750	4,700.080	11,029.830	11,029.830	11,029.830	11,502.046
01/02/2016-28/02/2016	4,445.350	3,394.720	7,840.070	7,840.070	7,840.070	8,040.060
01/03/2016-31/03/2016	6,266.050	4,950.120	11,216.170	11,216.170	11,216.170	11,072.030
01/04/2016-30/04/2016	6,717.900	5,052.880	11,770.780	11,770.780	11,770.780	12,252.160
01/05/2016-31/05/2016	5,243.000	3,844.400	9,087.400	9,087.400	9,087.400	9,165.824
01/06/2016-30/06/2016	5,472.600	4,028.920	9,501.520	9,501.520	9,501.520	9,898.518
01/07/2016-31/07/2016	3,997.000	2,877.560	6,874.560	6,874.560	6,874.560	8,034.838
01/08/2016-31/08/2016	3,522.400	2,393.160	5,915.560	5,915.560	5,915.560	5,798.384
01/09/2016-30/09/2016	3,993.150	3,061.800	7,054.950	7,054.950	7,054.950	7,454.226
01/10/2016-31/10/2016	4,307.800	3,166.240	7,474.040	7,474.040	7,474.040	7,742.656
01/11/2016-30/11/2016	6,022.100	4,545.800	10,567.900	10,567.900	10,567.900	11,180.952
01/12/2016-31/12/2016	5,990.600	4,446.400	10,437.000	10,437.000	10,437.000	11,209.678

Subtotal 2016	-	-	-	-	108,769.780	113,351.372
Total	-	-	-	-	579,392.800	604,211.890

Table 6: Electricity supplied by the project activity to the grid, Electricity imported from the grid by the project and Net electricity

supplied to the grid by the project activity

Monitoring period	EG _{export,project}				EG _{import,project}				EG _y
	Data based on meter readings M4	Data based on meter readings M5	Data based on meter readings M6	Subtotal	Data based on meter readings M4	Data based on meter readings M5	Data based on meter readings M6	Subtotal	Calculated by EG _{export,project} - EG _{import,project}
	MWh	MWh	MWh	MWh	MWh	MWh	MWh	MWh	MWh
	M	N	O	P=M+N+O	Q	R	S	P=M+N+O	U=P-T
26/10/2011-24/11/2011	2,915.052	4,772.488	2,261.560	9,949.100	9.912	16.184	8.092	34.188	9,914.912
25/11/2011-25/12/2011	3,335.332	5,821.144	2,158.212	11,314.688	8.428	13.300	7.028	28.756	11,285.932
26/12/2011-31/12/2011	632.016	1,033.844	506.744	2,172.604	1.260	1.988	1.064	4.312	2,168.292
Subtotal 2011	-	-	-	23,436.392	-	-	-	67.256	23,369.136
01/01/2012-25/01/2012	1,454.348	2,373.140	1,174.936	5,002.424	10.528	19.096	8.988	38.612	4,963.812
26/01/2012-24/02/2012	2,835.392	4,916.044	2,305.772	10,057.208	9.968	12.404	8.092	30.464	10,026.744
25/02/2012-25/03/2012	2,610.832	4,486.692	2,319.828	9,417.352	5.628	8.456	4.340	18.424	9,398.928
26/03/2012-24/04/2012	4,720.996	8,334.396	4,169.536	17,224.928	5.180	7.056	3.444	15.680	17,209.248
25/04/2012-25/05/2012	2,762.452	4,904.424	2,473.156	10,140.032	10.808	14.868	7.448	33.124	10,106.908
26/05/2012-30/06/2012	3,483.256	6,663.888	3034.948	13,182.092	8.932	13.776	6.860	29.568	13,152.524
01/07/2012-31/07/2012	1,604.456	3,922.548	1,711.220	7,238.224	9.744	18.200	9.716	37.660	7,200.564
01/08/2012-23/08/2012	914.900	1,664.264	761.180	3,340.344	11.508	17.052	9.212	37.772	3,302.572
24/08/2012-30/09/2012	2,820.160	4,776.576	2,132.872	9,729.608	11.312	17.836	9.520	38.668	9,690.940

01/10/2012-31/10/2012	2,871.484	5,091.212	2,436.280	10,398.976	6.216	9.800	4.144	20.160	10,378.816
01/11/2012-30/11/2012	3,851.792	6,986.056	3,082.212	13,920.060	4.956	7.280	3.752	15.988	13,904.072
01/12/2012-31/12/2012	3,763.564	6,190.408	2,989.308	12,943.280	5.432	9.016	4.760	19.208	12,924.072
Subtotal 2012	-	-	-	122,594.528	-	-	-	335.328	122,259.200
01/01/2013-31/01/2013	2,759.120	4,700.528	2,327.556	9,787.204	4.032	8.036	4.088	16.156	9,771.048
01/02/2013-28/02/2013	1,784.916	3,222.492	1,579.816	6,587.224	12.992	21.056	10.472	44.520	6,542.704
01/03/2013-31/03/2013	2,943.528	7,795.256	4,116.616	14,855.400	2.576	6.188	2.996	11.760	14,843.640
01/04/2013-30/04/2013	3,532.844	7,423.892	3,783.612	14,740.348	3.052	5.376	2.772	11.200	14,729.148
01/05/2013-31/05/2013	3,238.312	5,902.764	3,025.456	12,166.532	5.796	7.812	3.976	17.584	12,148.948
01/06/2013-30/06/2013	2,748.620	5,187.112	2,387.112	10,322.844	9.576	13.888	5.936	29.400	10,293.444
01/07/2013-31/07/2013	1,912.764	3,592.708	1,645.420	7,150.892	14.672	22.316	11.424	48.412	7,102.480
01/08/2013-31/08/2013	2,085.944	3,855.292	1,754.060	7,695.296	10.808	16.408	7.532	34.748	7,660.548
01/09/2013-30/09/2013	1,716.904	3,042.424	1,474.564	6,233.892	12.572	18.144	9.884	40.600	6,193.292
01/10/2013-31/10/2013	3,113.264	5,326.804	2,546.572	10,986.640	6.776	9.296	4.648	20.720	10,965.920
01/11/2013-30/11/2013	3,316.236	5,699.820	2,778.076	11,794.132	6.104	9.156	4.592	19.852	11,774.280
01/12/2013-31/12/2013	3,745.028	6,252.344	2,975.420	12,972.792	3.108	4.452	2.184	9.744	12,963.048
Subtotal 2013	-	-	-	125,293.196	-	-	-	304.696	124,988.500
01/01/2014-31/01/2014	2,925.580	5,157.600	2,649.780	10,732.960	3.920	5.432	2.884	12.236	10,720.724
01/02/2014-28/02/2014	2,846.396	5,099.780	2,639.728	10,585.904	6.216	9.408	5.208	20.832	10,565.072
01/03/2014-31/03/2014	2,911.776	5,364.660	2,800.812	11,077.248	6.636	9.996	5.068	21.700	11,055.548
01/04/2014-30/04/2014	2,656.528	4,629.464	2,401.448	9,687.440	8.764	12.320	6.244	27.328	9,660.112
01/05/2014-31/05/2014	4,172.728	7,564.284	3,759.868	15,496.880	4.704	6.328	2.884	13.916	15,482.964
01/06/2014-30/06/2014	1,581.412	2,890.216	1,458.688	5,930.316	11.452	17.528	8.708	37.688	5,892.628
01/07/2014-31/07/2014	2,225.832	4,095.644	1,916.124	8,237.600	10.892	16.240	8.484	35.616	8,201.984
01/08/2014-31/08/2014	926.212	1,643.236	801.444	3,370.892	15.708	23.940	11.984	51.632	3,319.260
01/09/2014-30/09/2014	1,556.352	2,581.628	1,349.096	5,487.076	8.876	13.972	7.588	30.436	5,456.640

01/10/2014-31/10/2014	3,077.284	5,411.224	2,741.368	11,229.876	2.000	12.712	6.524	21.236	11,208.640
01/11/2014-30/11/2014	2,455.600	3,929.968	2,074.268	8,459.836	6.636	9.548	5.292	21.476	8,438.360
01/12/2014-31/12/2014	4,517.408	6,672.120	4,027.156	15,216.684	3.808	5.516	2.688	12.012	15,204.672
Subtotal 2014	-	-	-	115,512.712	-	-	-	306.108	115,206.604
01/01/2015-31/01/2015	3,268.692	5,441.156	2,891.616	11,601.464	4.256	6.132	3.052	13.440	11,588.024
01/02/2015-28/02/2015	2,271.724	4,050.508	2,087.876	8,410.108	7.392	10.696	6.244	24.332	8,385.776
01/03/2015-31/03/2015	3,654.952	6,568.436	3,367.140	13,590.528	5.068	7.000	3.388	15.456	13,575.072
01/04/2015-30/04/2015	3,857.700	6,796.384	3,509.296	14,163.380	6.832	8.932	4.620	20.384	14,142.996
01/05/2015-31/05/2015	3,377.304	5,617.836	3,091.088	12,086.228	7.924	9.772	5.292	22.988	12,063.240
01/06/2015-30/06/2015	2,234.288	3,919.692	2,036.580	8,190.560	11.060	15.176	8.708	34.944	8,155.616
01/07/2015-31/07/2015	1,615.740	2,673.244	1,386.308	5,675.292	11.088	15.736	8.568	35.392	5,639.900
01/08/2015-31/08/2015	868.756	1,425.060	771.232	3,065.048	14.784	19.936	11.564	46.284	3,018.764
01/09/2015-30/09/2015	1,206.072	1,870.484	1,018.052	4,094.608	17.360	23.968	14.140	55.468	4,039.140
01/10/2015-31/10/2015	2,865.016	4,237.016	2,564.156	9,666.188	6.300	8.792	4.956	20.048	9,646.140
01/11/2015-30/11/2015	3,575.684	5,464.872	2,944.564	11,985.120	5.600	8.456	4.620	18.676	11,966.444
01/12/2015-31/12/2015	2,962.372	5,585.608	2,753.380	11,301.360	8.008	11.928	6.412	26.348	11,275.012
Subtotal 2015	-	-	-	113,829.884	-	-	-	333.760	113,496.124
01/01/2016-31/01/2016	3,247.328	5,635.280	2,857.596	11,740.204	4.760	7.476	3.640	15.876	11,724.328
01/02/2016-28/02/2016	2,314.368	4,303.292	2,056.348	8,674.008	7.168	10.556	5.740	23.464	8,650.544
01/03/2016-31/03/2016	3,114.496	5,513.648	2,957.948	11,586.092	3.920	4.424	2.100	10.444	11,575.648
01/04/2016-30/04/2016	3,316.124	5,883.752	2,992.248	12,192.124	4.704	6.412	3.080	14.196	12,177.928
01/05/2016-31/05/2016	2,545.088	4,680.676	2,366.700	9,592.464	6.664	9.016	4.256	19.936	9,572.528
01/06/2016-30/06/2016	2,738.988	4,991.028	2,443.588	10,173.604	7.840	11.004	5.292	24.136	10,149.468
01/07/2016-31/07/2016	2,036.076	3,676.260	1,656.788	7,369.124	12.068	18.340	9.296	39.704	7,329.420
01/08/2016-31/08/2016	1,763.328	3,041.640	1,475.124	6,280.092	10.976	15.540	7.756	34.272	6,245.820
01/09/2016-30/09/2016	2,012.696	3,810.044	1,907.528	7,730.268	12.432	19.180	9.912	41.524	7,688.744

01/10/2016-31/10/2016	2,238.208	3,757.852	1,972.544	7,968.604	6.076	8.428	5.040	19.544	7,949.060
01/11/2016-30/11/2016	3,158.876	5,493.124	2,744.084	11,396.084	4.648	6.244	3.472	14.364	11,381.720
01/12/2016-31/12/2016	3,287.172	5,370.232	2,736.076	11,393.480	5.936	8.876	4.452	19.264	11,374.216
Subtotal 2016	-	-	-	116,096.148	-	-	-	276.724	115,819.424
Total	-	-	-	616,762.860	-	-	-	1,623.872	615,138.988

Table 7: Net electricity supplied to the grid by the project activity

Monitoring period	EG _y		
	Calculated by $EG_{\text{export, total}} - EG_{\text{import, total}} - EG_{\text{export, other, y}}$	Calculated by $EG_{\text{export, project}} - EG_{\text{import, project}}$	Conservative value
	Unit: MWh	Unit: MWh	Unit: MWh
	L	U	W=Min(L,U)
26/10/2011-24/11/2011	9,407.290	9,914.912	9,407.290
25/11/2011-25/12/2011	12,033.680	11,285.932	11,285.932
26/12/2011-31/12/2011	1,838.650	2,168.292	1,838.650
Subtotal 2011	23,279.620	23,369.136	22,531.872
01/01/2012-25/01/2012	5,043.800	4,963.812	4,963.812
26/01/2012-24/02/2012	9,602.130	10,026.744	9,602.130
25/02/2012-25/03/2012	10,016.890	9,398.928	9,398.928
26/03/2012-24/04/2012	15,734.330	17,209.248	15,734.330
25/04/2012-25/05/2012	9,934.400	10,106.908	9,934.400
26/05/2012-30/06/2012	12,969.920	13,152.524	12,969.920
01/07/2012-31/07/2012	7,593.560	7,200.564	7,200.564
01/08/2012-23/08/2012	3,198.220	3,302.572	3,198.220
24/08/2012-30/09/2012	9,454.440	9,690.940	9,454.440

01/10/2012-31/10/2012	10,171.700	10,378.816	10,171.700
01/11/2012-30/11/2012	13,671.430	13,904.072	13,671.430
01/12/2012-31/12/2012	12,674.810	12,924.072	12,674.810
Subtotal 2012	120,065.630	122,259.200	118,974.684
01/01/2013-31/01/2013	9,598.690	9,771.048	9,598.690
01/02/2013-28/02/2013	6,301.990	6,542.704	6,301.990
01/03/2013-31/03/2013	16,034.320	14,843.640	14,843.640
01/04/2013-30/04/2013	15,161.330	14,729.148	14,729.148
01/05/2013-31/05/2013	11,941.140	12,148.948	11,941.140
01/06/2013-30/06/2013	10,070.090	10,293.444	10,070.090
01/07/2013-31/07/2013	6,912.140	7,102.480	6,912.140
01/08/2013-31/08/2013	7,471.170	7,660.548	7,471.170
01/09/2013-30/09/2013	6,120.370	6,193.292	6,120.370
01/10/2013-31/10/2013	10,619.830	10,965.920	10,619.830
01/11/2013-30/11/2013	11,686.750	11,774.280	11,686.750
01/12/2013-31/12/2013	12,621.890	12,963.048	12,621.890
Subtotal 2013	124,539.710	124,988.500	122,916.848
01/01/2014-31/01/2014	10,292.430	10720.724	10,292.430
01/02/2014-28/02/2014	10,517.380	10565.072	10,517.380
01/03/2014-31/03/2014	10,545.690	11055.548	10,545.690
01/04/2014-30/04/2014	9,760.160	9660.112	9,660.112
01/05/2014-31/05/2014	15,640.420	15482.964	15,482.964
01/06/2014-30/06/2014	5,395.240	5892.628	5,395.240
01/07/2014-31/07/2014	8,007.930	8201.984	8,007.930
01/08/2014-31/08/2014	3,018.260	3319.260	3,018.260
01/09/2014-30/09/2014	4,626.910	5456.640	4,626.910

01/10/2014-31/10/2014	11,546.260	11,208.640	11,208.640
01/11/2014-30/11/2014	7,901.230	8,438.360	7,901.230
01/12/2014-31/12/2014	15,288.510	15,204.672	15,204.672
Subtotal 2014	112,540.420	115,206.604	111,861.458
01/01/2015-31/01/2015	11,828.680	11,588.024	11,588.024
01/02/2015-28/02/2015	7,993.160	8,385.776	7,993.160
01/03/2015-31/03/2015	13,339.540	13,575.072	13,339.540
01/04/2015-30/04/2015	13,697.950	14,142.996	13,697.950
01/05/2015-31/05/2015	11,447.530	12,063.240	11,447.530
01/06/2015-30/06/2015	8,514.206	8,155.616	8,155.616
01/07/2015-31/07/2015	5,363.414	5,639.900	5,363.414
01/08/2015-31/08/2015	2,729.730	3,018.764	2,729.730
01/09/2015-30/09/2015	4,136.954	4,039.140	4,039.140
01/10/2015-31/10/2015	8,835.422	9,646.140	8,835.422
01/11/2015-30/11/2015	11,582.434	11,966.444	11,582.434
01/12/2015-31/12/2015	10,966.118	11,275.012	10,966.118
Subtotal 2015	110,435.138	113,496.124	109,738.078
01/01/2016-31/01/2016	11,502.046	11,724.328	11,502.046
01/02/2016-28/02/2016	8,040.060	8,650.544	8,040.060
01/03/2016-31/03/2016	11,072.030	11,575.648	11,072.030
01/04/2016-30/04/2016	12,252.160	12,177.928	12,177.928
01/05/2016-31/05/2016	9,165.824	9,572.528	9,165.824
01/06/2016-30/06/2016	9,898.518	10,149.468	9,898.518
01/07/2016-31/07/2016	7,407.838	7,329.420	7,329.420
01/08/2016-31/08/2016	5,798.384	6,245.820	5,798.384
01/09/2016-30/09/2016	7,454.226	7,688.744	7,454.226

01/10/2016-31/10/2016	7,742.656	7,949.060	7,742.656
01/11/2016-30/11/2016	11,180.952	11,381.720	11,180.952
01/12/2016-31/12/2016	11,209.678	11,374.216	11,209.678
Subtotal 2016	113,351.372	115,819.424	112,571.722
Total	604,211.890	615,138.988	598,594.662

APPENDIX 2: < EVIDENCE OF ACHIEVED SD CONTRIBUTIONS >

SDG Target 7.2.1 Renewable energy share in the total final energy and SDG Target 13 Tonnes of greenhouse gas emissions avoided or removed in the total final energy consumption

Monitoring period	Amount of renewable energy (MWh)	Emission reductions (tCO ₂ e)	Data source
20/12/2008-20/07/2009	67,622.300	71,328	https://registry.verra.org/app/search/VCS Amount of renewable energy was calculated from emission reduction which is posted on the above website.
26/10/2011-31/12/2016	598,594.662	631,397	Section 5 of this report
Current project contributions	666,216.962	702,725	Calculated including both the contributions during the first monitoring period (20/12/2008-20/07/2009) and this monitoring period (26/10/2011-31/12/2016) under VCS.
Contributions over project lifetime	666,216.962	702,725	Calculated including both the contributions during the first monitoring period (20/12/2008-20/07/2009) and this monitoring period (26/10/2011-31/12/2016) under VCS.

SDG Target 8.5 Number of jobs created

华润电力风能（烟台）有限公司-蓬莱平顶山风电场

职工花名册（2016 年）

序号	姓名	职位	性别
1	王培德	场长	男
2	张金鑫	助理场长	男
3	李杰	检修巡操岗	男
4	周伟	检修巡操岗	男
5	仝城	检修巡操岗	男
6	杨峰	检修巡操岗	男
7	李鹏建	检修巡操岗	男
8	张明	检修巡操岗	男
9	张焕刚	检修巡操岗	男
10	古海光	检修巡操岗	男
11	王磊	检修巡操岗	男
12	黄鹏杰	检修巡操岗	男
13	于永琪	检修巡操岗	男
14	张琰	财务岗	女
15	邹贡哲	检修巡操岗	男
16	邱剑	检修巡操岗	男
17	丁洋	安全管理岗	男
18	吴龙	检修巡操岗	男
19	倪建伟	检修巡操岗	男
20	王飞	检修巡操岗	男
21	徐嘉	人力资源岗（兼职）	女
22	李晨阳	会计岗（兼职）	女
23	郭田田	会计岗（兼职）	女
24	朱文波	商务岗（兼职）	女